Deriving Diffusion Coefficient D via Numerical PDE Inversion

Full Mathematical Framework for Comparing Oxygen Delivery Materials

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1. The Physical Problem

You have an oxygen-carrier material (e.g. PFCMC) loaded with O_2 at its core.
Over time, oxygen diffuses outward through the material cross-section.
You measure the oxygen concentration profile $C(x, t)$ along a 1D diagonal section
at discrete positions $x \in \{x_1, ..., x_N\}$ and times $t \in \{t_0, t_1, ..., t_M\}$,
where $\Delta t = 5$ min.

Goal: Extract D (mm^2/s) and a retention metric to rank materials.

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2. Governing Equation

Fick's Second Law of Diffusion in 1D:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} \tag{1}$$

This is a linear parabolic PDE. It says the *rate of change* of concentration at any point equals D times the *curvature* of the concentration profile at that point.

- Where the profile is concave down ($\partial^2 C/\partial x^2 < 0$, i.e. at the peak), C decreases.
- Where the profile is concave up ($\partial^2 C/\partial x^2 > 0$, i.e. at the edges), C increases.
- The rate of all this is governed by D.

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3. Discretisation: Crank-Nicolson Method

3.1 Why Crank-Nicolson?

Three options exist for discretising Eq.(1) in time:

Method	Scheme	Stability	Accuracy
Explicit (Forward Euler)	C^(n+1) from C^n directly	Requires $\Delta t < \Delta x^2/(2D)$ — **unstable for your data**	$O(\Delta t, \Delta x^2)$
Implicit (Backward Euler)	Solve linear system	Unconditionally stable	$O(\Delta t, \Delta x^2)$
Crank-Nicolson	**Average of explicit + implicit**	**Unconditionally stable**	**$O(\Delta t^2, \Delta x^2)$**

Your data has $\Delta x = 0.0345$ mm and $\Delta t = 300$ s. For $D \approx 10^{-3}$ mm²/s, the explicit stability limit would be $\Delta t < (0.0345)^2/(2 \times 10^{-3}) \approx 0.6$ s — you'd need 500 sub-steps per measurement interval. Crank-Nicolson has no such restriction AND is second-order accurate in time.

3.2 The Scheme

Evaluate the spatial derivative as the average of timesteps n and n+1:

$$
\frac{C_i^{n+1} - C_i^n}{\Delta t}
= \frac{D}{2} \left[
\frac{C_{i-1}^{n+1} - 2C_i^{n+1} + C_{i+1}^{n+1}}{\Delta x^2}

- \frac{C_{i-1}^{n} - 2C_i^{n} + C_{i+1}^{n}}{\Delta x^2} \right] \tag{2} $$

Define the mesh ratio:

$$r = \frac{D\,\Delta t}{2\,\Delta x^2} \tag{3}$$

Rearranging (2), group unknowns (n+1) on the left, knowns (n) on the right:

$$-r\,C_{i-1}^{n+1} + (1+2r)\,C_i^{n+1} - r\,C_{i+1}^{n+1} = r\,C_{i-1}^{n} + (1-2r)\,C_i^{n} + r\,C_{i+1}^{n} \tag{4}$$

3.3 Matrix Form

For interior points i = 1, 2, ..., N-2 (with i=0 and i=N-1 as boundaries), this is a tridiagonal linear system:

$$\mathbf{A}\,\mathbf{C}^{n+1} = \mathbf{B}\,\mathbf{C}^{n} + \mathbf{b}^{n} \tag{5}$$

where:

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        [ 1+2r   -r                ]         [ 1-2r    r                ]
        |  -r   1+2r  -r           |         |   r   1-2r   r           |
  A  =  |        -r  1+2r  -r      |,   B =  |         r  1-2r   r      |
        |              ...         |         |              ...         |
        [             -r  1+2r     ]         [              r  1-2r     ]
```

The vector $\mathbf{b}^n$ accounts for boundary conditions (known values at edges).

3.4 Boundary Conditions

Dirichlet (fixed-value) BCs — we pin the edges to measured values:

$$C(x_0, t) = C_{\text{left}}(t), \quad C(x_{N-1}, t) = C_{\text{right}}(t) \tag{6}$$

In practice, we use the average of the first/last 10 pixels (smoothed edges) to reduce noise. This is the correct physical BC because your material exchanges O_2 with the surrounding environment at its boundaries.

3.5 Thomas Algorithm (Tridiagonal Solve)

The tridiagonal system (5) is solved in O(N) operations via forward elimination and back-substitution:

Forward sweep (i = 1, ..., N-2):

$$c'_i = \frac{c_i}{b_i - a_i\, c'_{i-1}}, \quad d'_i = \frac{d_i - a_i\, d'_{i-1}}{b_i - a_i\, c'_{i-1}} \tag{7}$$

Back-substitution (i = N-3, ..., 0):

$$C_i^{n+1} = d'_i - c'_i\, C_{i+1}^{n+1} \tag{8}$$

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4. The Inverse Problem: Finding D

4.1 Forward Problem

Given D, we can compute C_sim(x, t; D) by marching Eq.(4) forward from the initial condition $C(x, t_0)$ = measured first profile.

4.2 Objective Function

Define the misfit between simulation and data:

$$J(D) = \sum_{n=1}^{M} \sum_{i=m}^{N-m} \left[C_{\text{sim}}(x_i, t_n; D) - C_{\text{data}}(x_i, t_n)\right]^2 \tag{9}$$

where m is a margin (we exclude edge pixels to avoid BC bias).

4.3 Optimisation

D is a single scalar. The objective J(D) is smooth and unimodal (verified by scanning D over orders of magnitude). We minimise with bounded scalar optimisation:

$$D^* = \underset{D \in [10^{-6},\, 5\times10^{-2}]}{\arg\min} \; J(D) \tag{10}$$

Using `scipy.optimize.minimize_scalar` with Brent's method (bounded).

4.4 Quality Metric

$$\text{RMSE} = \sqrt{\frac{J(D^*)}{(N-2m)(M-1)}} \tag{11}$$

This is in the same units as C (% Air Saturation), directly interpretable as "how well does simple Fickian diffusion explain your data."

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5. Why the PDE Method is Superior for Material Comparison

5.1 The Moment Method's Limitations for Your Data

The moment method (D = slope(σ^2)/2) assumes:

1. **Unbounded domain** — your material is finite (12.5 mm). Once the diffusion front hits the edges, σ^2 saturates and the method fails. In your data, σ^2 plateaus after ~100 min.
2. **Conserved total mass** — your data shows M_0 dropping from 414 to ~180 (O_2 leaves through boundaries). The variance of a shrinking distribution is not governed by $d\sigma^2/dt = 2D$ alone.
3. **Positive excess everywhere** — at late times, edge concentrations exceed the initial baseline, making "excess" negative at boundaries. This corrupts the moment integrals.
4. **Only uses early data** — restricted to the first ~60 min out of 485 min of data. Throws away 88% of your measurements.

5.2 The PDE Method's Advantages

Property	Moment Method	PDE Method
Uses all timesteps	No (early only)	**Yes (all 98)**
Handles finite domains	No	**Yes (Dirichlet BCs)**
Handles O_2 loss at edges	No	**Yes (measured BCs)**
Handles late-time flattening	No	**Yes**
Assumes profile shape	No (good)	No (good)
Model-free	Yes	No (assumes Fick's law)
Output	Single number	**D + full predicted C(x,t)**
Residual diagnostics	R^2 on σ^2 only	**Per-pixel, per-timestep RMSE**

5.3 Why This Matters for Comparing Materials

When you compare Material A vs Material B, you need a D that is:

1. **Consistent across conditions**: The PDE method uses ALL your data, so it averages over noise and gives a single robust D even if individual timesteps are noisy.
2. **Physically correct at boundaries**: Your materials exchange O_2 with the environment. Ignoring this (as the moment method does) systematically biases D.
3. **Accompanied by a goodness-of-fit**: The RMSE tells you whether Fick's law even applies. If Material A has RMSE = 5 and Material B has RMSE = 25, then Material B may have concentration-dependent D or reaction kinetics — important to know!
4. **Reproducible**: The PDE method has no subjective choices about "how many early timesteps to include." Same data → same D, always.

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6. The Retention Metric: τ_half

For ranking "which material stays oxygenated longest," define:

$$C_{\text{peak}}^{\text{ex}}(t) = \max_x \left[C(x,t) - C_\infty\right] \tag{12}$$

$$\tau_{1/2} = \text{time at which} \quad C_{\text{peak}}^{\text{ex}}(\tau_{1/2}) = \frac{1}{2}\, C_{\text{peak}}^{\text{ex}}(0) \tag{13}$$

$\tau_{1/2}$ is the half-life of the peak oxygen excess.

6.1 Why $\tau_{1/2}$ and Not Just 1/D?

For identical geometry, $\tau_{1/2} \propto 1/D$, so lower D means longer retention.
But $\tau_{1/2}$ also depends on:

- **Initial loading width** (σ_0): A broader initial loading decays slower even with the same D.
- **Material thickness**: Thicker samples retain longer.
- **Boundary conditions**: Materials with less permeable surfaces retain longer.

$\tau_{1/2}$ captures ALL of these effects in a single number. D alone does not.

6.2 For an Ideal Gaussian in Unbounded Medium

If C(x,0) is Gaussian with variance σ_0^2, the peak decays as:

$$C_{\text{peak}}(t) \propto \frac{1}{\sqrt{\sigma_0^2 + 2Dt}} \tag{14}$$

Setting this to half its initial value:

$$\frac{1}{\sqrt{\sigma_0^2 + 2D\tau}} = \frac{1}{2} \cdot \frac{1}{\sqrt{\sigma_0^2}}$$

$$\tau_{1/2} = \frac{3\,\sigma_0^2}{2\,D} \tag{15}$$

But your data is NOT an ideal Gaussian in an unbounded medium. That's why we measure $\tau_{1/2}$ directly from the data rather than computing it from Eq.(15).

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7. Complete Workflow for Material Comparison

For each material sample:

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1. Acquire C(x, t) cross-section data at 5-min intervals
2. Detect loading→diffusion transition (auto: peak of centre-region mean)
3. Trim loading phase
4. Run PDE inversion → D*, RMSE
5. Measure τ1/2 from peak excess decay
6. Record: D*, RMSE, τ1/2, initial peak, FWHM0
```

Comparison table (example):

Material	D (mm^2/s)	RMSE	$\tau_{1/2}$ (min)	$FWHM_0$ (mm)	Verdict
PFCMC-A	5.9×10^{-4}	3.2	85	4.3	Best retention
PFCMC-B	1.2×10^{-3}	4.1	52	4.1	Moderate
Hydrogel-C	3.5×10^{-3}	5.8	18	3.8	Fast release

Primary ranking metric: $\tau_{1/2}$ (directly answers "stays oxygenated longest")

Secondary: D (intrinsic material property, geometry-independent)

Quality check: RMSE (if high → Fick's law may not apply → flag for further investigation)

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8. Summary of Key Equations

Quantity	Equation	Number
Fick's Law	$\partial C/\partial t = D\, \partial^2 C/\partial x^2$	(1)
Crank-Nicolson stencil	$-r\, C_{i-1}^{n+1} + (1+2r)\, C_i^{n+1} - r\, C_{i+1}^{n+1} = RHS$	(4)
Mesh ratio	$r = D\, \Delta t / (2\, \Delta x^2)$	(3)
Objective	$J(D) = \Sigma\, [C_{sim} - C_{data}]^2$	(9)
Optimal D	$D^* = \mathrm{argmin}\, J(D)$	(10)
Retention half-life	$\tau_{1/2}$: time when peak excess halves	(13)

The Python script `diffusion_D.py` *implements all of the above.*